

Stateflow Modelling Based Transistor Sizing for Optimised Digital Circuit Performance

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What we solve?

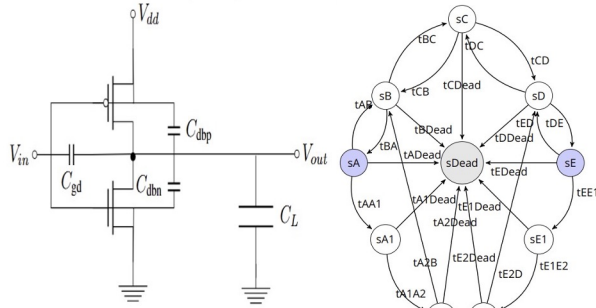
Modelling of CMOS circuits as a hybrid system at SPICE Level2.

Specification and verification of desired circuit behaviour using STL.

Generate multiple feasible transistor sizing solutions using a proposed 2D search algorithm.

Why does it matter?

- Automatically identifies feasible transistor parameter ranges.
- Detects incomplete models and unreachable (dead) states early in the design process.
- Improve reliability, design productivity, and automation in CMOS circuit design.

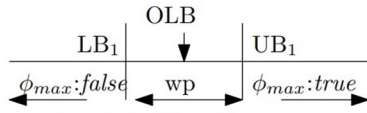


Loc	pMOS	nMOS
sA	Triode	Cut-off
sA1	Triode	Cut-off
sA2	Triode	Cut-off
sB	Triode	Saturation
sC	Saturation	Saturation
sD	Saturation	Triode
sE	Cut-off	Triode
sE1	Cut-off	Triode
sE2	Cut-off	Triode
sF	Saturation	Cut-off
sG	Cut-off	Saturation
sH	Cut-off	Cut-off
sI	Triode	Triode
sDead	-	-

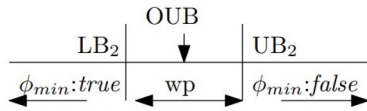
Key Idea

- Breach [1] estimates reachable sets of the state model
- Breach returns FALSE (0) if no satisfying trace exists for STL property; else return TRUE (1)
- Monotonicity of STL properties wrt transistor sizing

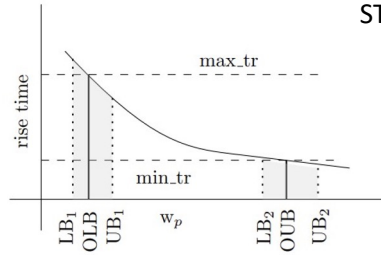
Methodology to find the optimal range for performance requirements



(a) LB₁, UB₁ by Algorithm



(b) LB₂, UB₂ by Algorithm

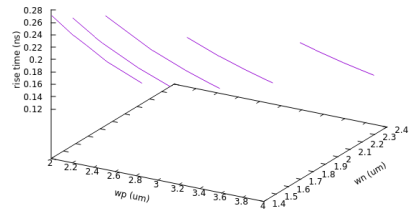


(c) Working of Algorithm

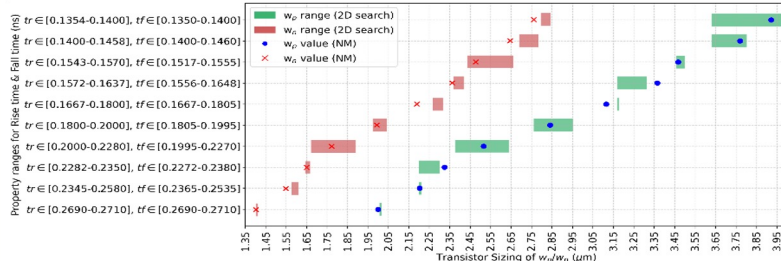
STL Properties to find timing range

$$\phi_{\min} : G_{[0.8e-9, 1e-9]}(t_f[t] > \min_tr \wedge t_f[t] > \min_tf)$$

$$\phi_{\max} : G_{[0.8e-9, 1e-9]}(t_f[t] < \max_tr \wedge t_f[t] < \max_tf)$$



Experimental results



- Timing optimisation on CMOS inverter using NM[2], 2D search
- Paired T-test (SPICE[3] and 2D search results): not statistically significant difference for rise time ($t = 0.67$, $p = 0.518$), fall time ($t = 1.377$, $p = 0.202$)

Conclusion

- CMOS circuits were successfully modelled as hybrid systems.
- STL-based performance enabled automated parameter extraction for transistor sizing, by optimising timing and power.
- The proposed 2D search algorithm generated set of feasible design solutions based on results for CMOS inverter and 2-input NAND and NOR

[1] A. Donzé, "Breach: A toolbox for verification and parameter synthesis of hybrid systems," Intl. Conf. on CAV, pp. 167–170, 2010.

[2] J. A. Nelder and R. Mead, "A simplex method for function minimisation", The Computer Journal, vol. 7, pp. 308–313, 1965.

[3] AIM-Spice Reference Manual. 2014.